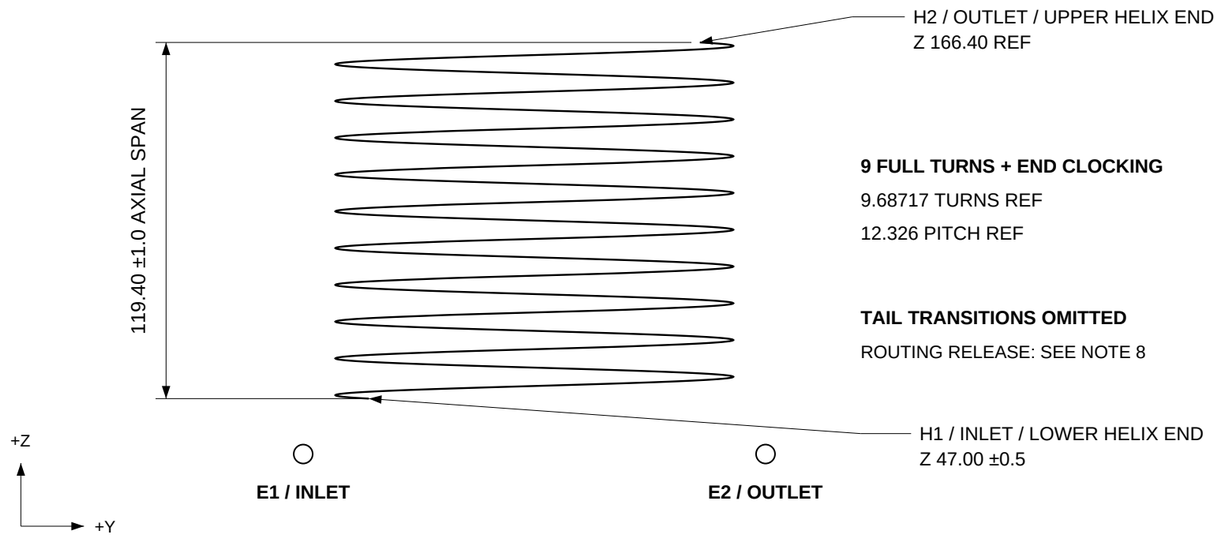
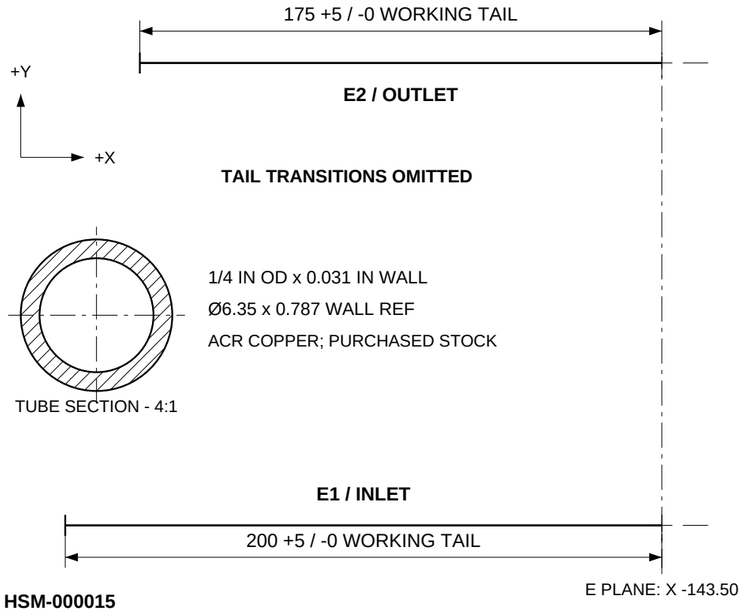
INSTALLED ELEVATION - LOOKING ALONG -X

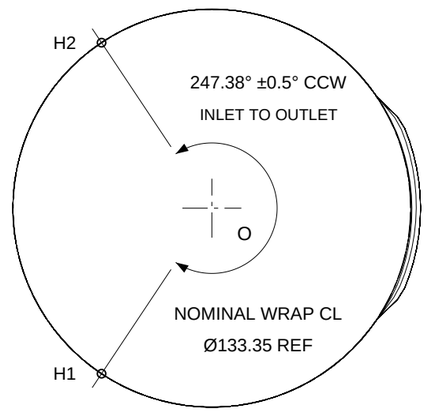
TUBE CENTERLINES SHOWN - SCALE 1:2



PLAN - LOOKING ALONG -Z

TAIL CENTERLINES / WRAP ENVELOPE - SCALE 1:2





INSTALLED CENTERLINE COORDINATES

COLD-CORE NATIVE FRAME; SEE NOTE 1

POINT	X	Y	Z	FUNCTION
H1	-36.98	-55.48	47.000	HELIX START
H2	-36.98	55.48	166.400	HELIX END
E1/E2	-143.50	-77.50 / +77.50	28.425	EXITS

E1 = -Y / INLET; E2 = +Y / OUTLET. EXIT Y AND Z: ±0.25.

H1/H2 X,Y ARE REF; SPAN / CLOCKING CONTROL THE WRAP.

NOMINAL SHELL PORT STATIONS: X -141.50 REF, 2 mm INBOARD OF E.

FORMING AND FINISHED-STATE REQUIREMENTS

1. One continuous length of GOORY 1/4-inch OD x 0.031-inch wall ACR copper. O is the carbonator axis in X,Y; Z=0 is the cold-core shell base. Dimensions describe the coil fitted to the vessel and bridge, before foam and final refrigeration tie-in.

2. Cut stock 4854 +10/-0. Nominal laid wrap 4088.5 REF plus inlet/outlet allowances 335.5/429.6 REF. Preserve each working tail's stated length at E; trim only during refrigeration fit-up.

3. Use the Ø123 winding mandrel as the forming aid; adjust winding and tails to the INSTALLED dimensions. The mandrel is not the acceptance gauge. Keep nine complete turns plus the stated CCW end clocking, rising from inlet to outlet. Pitch is derived from axial span and turns.

4. Fit the nominal Ø127 vessel with foil thermal interface; maintain contact around the available wrap. Ease locally over the reed bridge: 3.25 REF radial lift. Do not compress reeds or force the copper through the bridge. Free-state diameter / springback is not a controlled finished dimension.

5. Blend helix-to-tail transitions smoothly; all tail bends R12.7 MIN to tube centerline. Both exit axes parallel to -X within 1°. Exit centers at E1/E2 within the table limits; confirm passage through the actual shell feed-throughs before foam.

6. Finished tube must have an open bore, no kink, crease or visible flattening. Square-cut and deburr both open ends; keep the bore clean and capped until tie-in. Retain clearance to reservoirs, reeds and other lines in the assembled core. Inspect before foam conceals the coil.

7. The working tails belong to the refrigeration fit-up; braze joints, heat protection, leak checks and final charge follow assembly/refrigerant-loop.md. Confirm forming capability and installed fit on the first article.

8. TAIL ROUTE RELEASE ITEM: current layout arcs reach R5.72 inlet / R8.78 outlet REF, below the required minimum. Intermediate routes are omitted. Establish compliant transitions and assembled clearances, then verify a formed first article before release.

HSM-000015

mm DIMENSIONS GOVERN.

DO NOT SCALE.

REF = REFERENCE.

PRINT AT 100%
FOR STATED SCALES.

HOME SODA MACHINE **EVAPORATOR COIL**				**FORMING DEFINITION** TAIL ROUTE PENDING
MATERIAL **ACR COPPER TUBE**		UNITS **MILLIMETERS**	QUANTITY **1 PER COLD CORE**	
DRAWING **EVAP-01**	REV **A**	DATE **2026-10-27**	SCALE **AS SHOWN**	SHEET **1 OF 1**